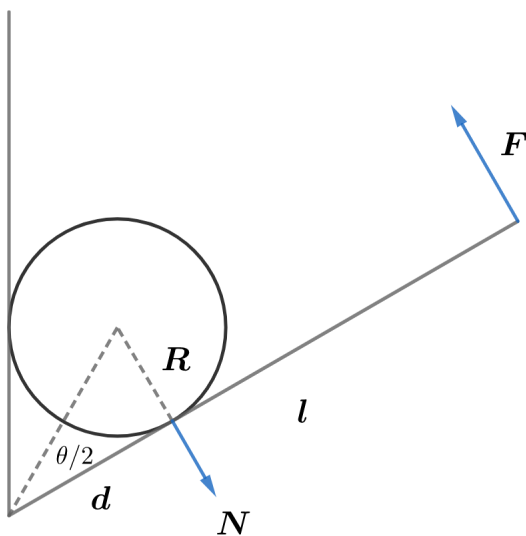


2019B F=ma Exam: Problem 9

Kevin S. Huang



Balancing torques on the board with our pivot at the hinge, we have

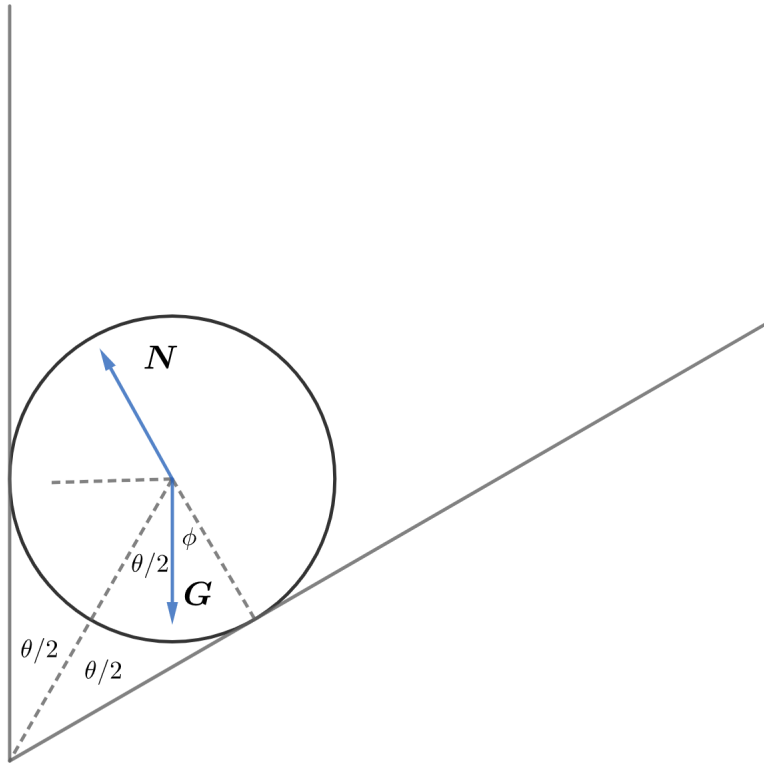
$$Fl = Nd$$

Note that

$$\tan \frac{\theta}{2} = \tan \frac{\pi}{6} = \frac{\sqrt{3}}{3} = \frac{R}{d}$$

so

$$d = R\sqrt{3}$$



The vertical component of N must balance G so

$$N \cos \phi = G$$

Since

$$\theta + \phi = \frac{\pi}{2}$$

$$\phi = \frac{\pi}{6}$$

we have

$$N = \frac{2G}{\sqrt{3}}$$

Hence,

$$F = \frac{Nd}{l} = \frac{2G}{\sqrt{3}} \frac{R\sqrt{3}}{l} = \frac{2GR}{l}$$

so the answer is B.